

GR5: Low-Fi Prototyping

Daniil Novak, Ved Pradhan, Rishik Buneti



The Team



Daniil Novak

'28
Santa Barbara, CA



Ved Pradhan

'26
Santa Clara, CA



Rishik Buneti

'26
Cupertino, CA

Problem / Solution Overview

College students and Gen Z shoppers often turn to online shopping as an emotional activity, but current platforms feel impersonal and overwhelming. This app introduces an AI shopping companion that holds a natural, supportive conversation with the user to understand their mood, context, and intent. Based on that interaction, the AI curates and presents products in a way that feels personal, low-pressure, and emotionally aligned.





TEXTURE

Product discovery that knows how you feel



Outline

1. Concept Sketches
2. Key Screen Sketches
3. Deciding on a Modality
4. Task Flows
5. Low-Fi Prototyping
6. Testing Methodology
7. Testing Results
8. Appendix





Concept Sketches

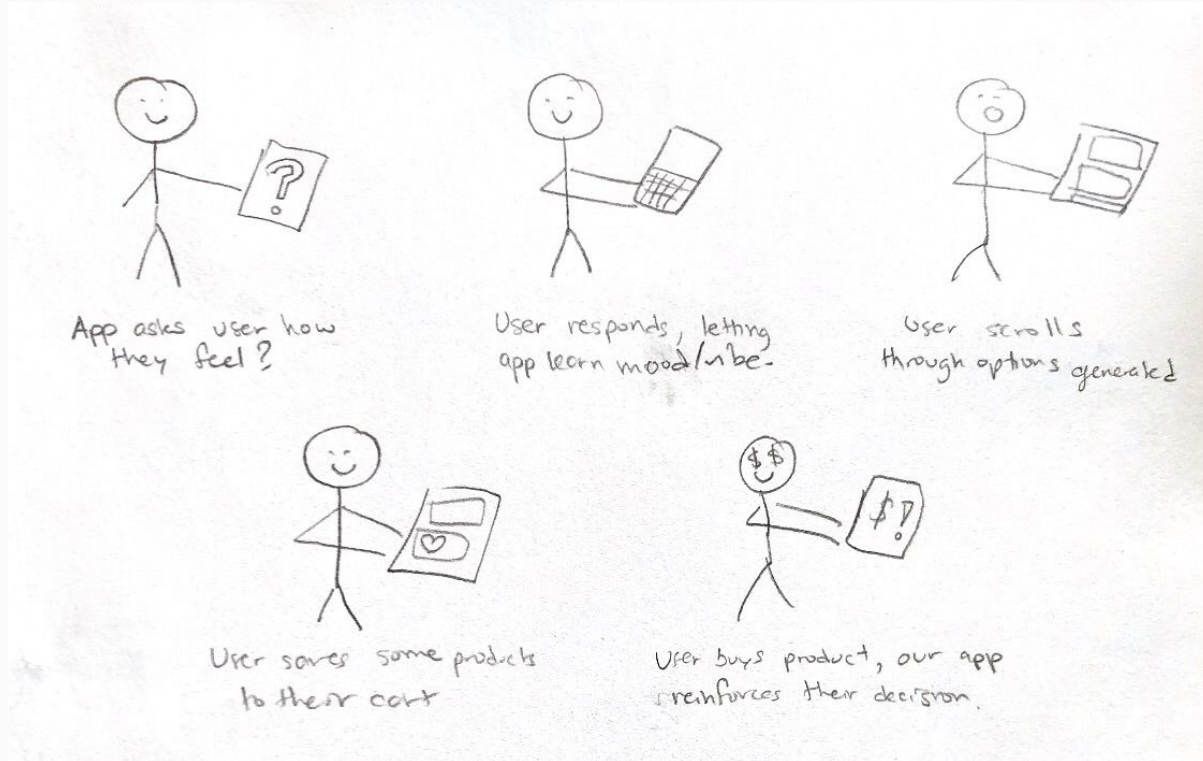


Modality Brainstorming

- Barebones Phone App
- Google Home integration
- AR Glasses
- Smartwatch
- Desktop application
- Responsive web application
- Dedicated Tablet
- Smart mirror
- Dedicated Kiosks at stores
- Physical Humanoid Robot



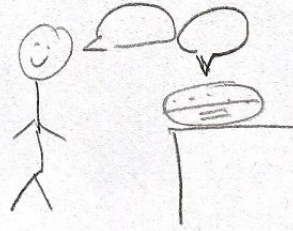
Barebones Phone App



Google Home / Alexa Integration



User starts conversation with Google Home



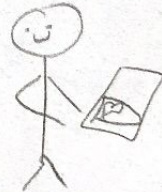
Google Home asks user how they feel and records mood



Home asynchronously sends product recommendations to user's phone.



User checks phone for notifications from Google Home.



User selects what they like, Google Home can give more guidance if needed.

AR Glasses



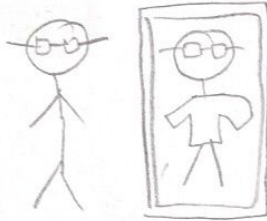
Glasses initiates conversation with user when detects store entrance



Glasses learns user's vibe/mood.



Glasses scans environment; points out best-match products.



In front of mirror, glasses can show what the user will look like.



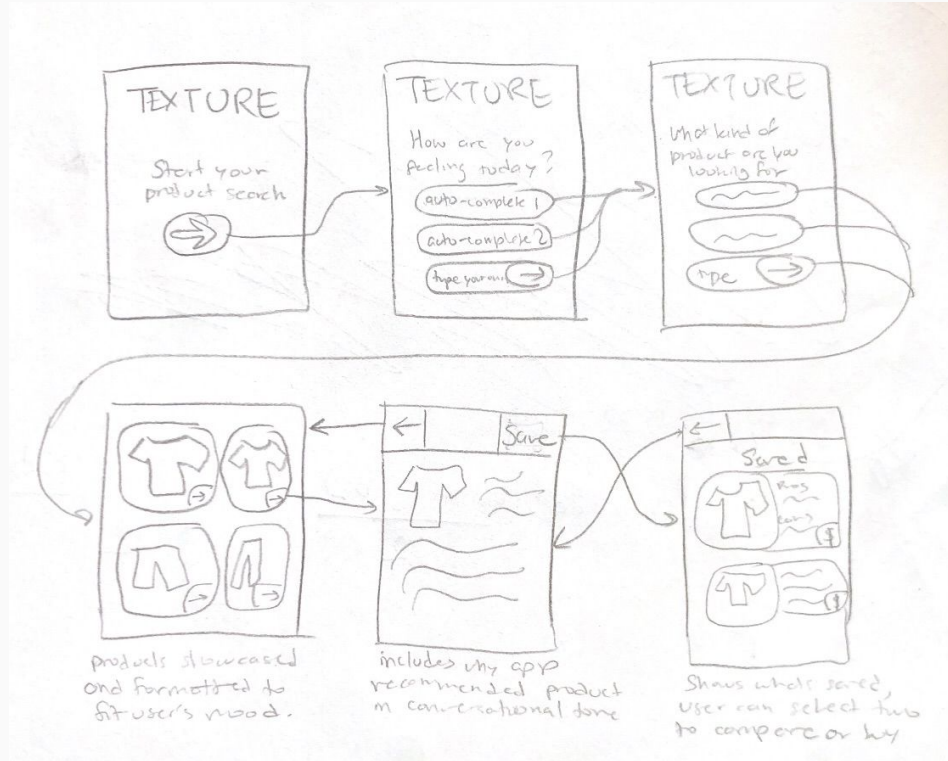
Glasses reinforce user's decision to buy product.



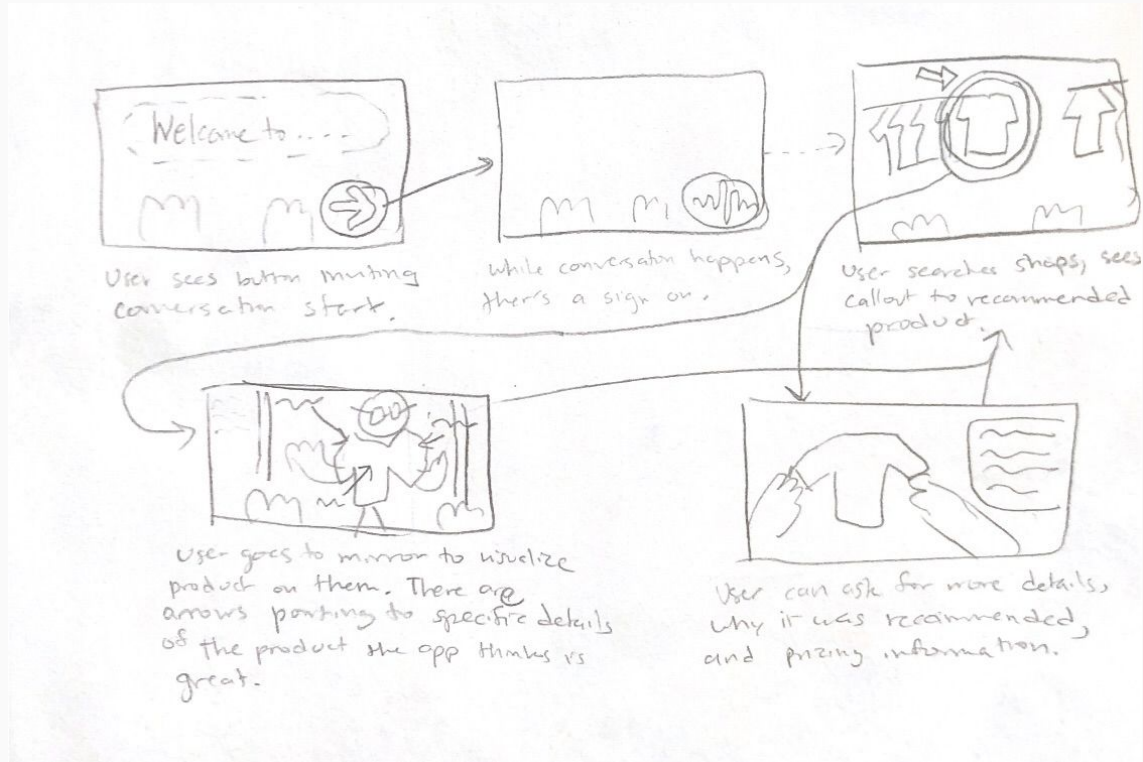
Key Screen Sketches

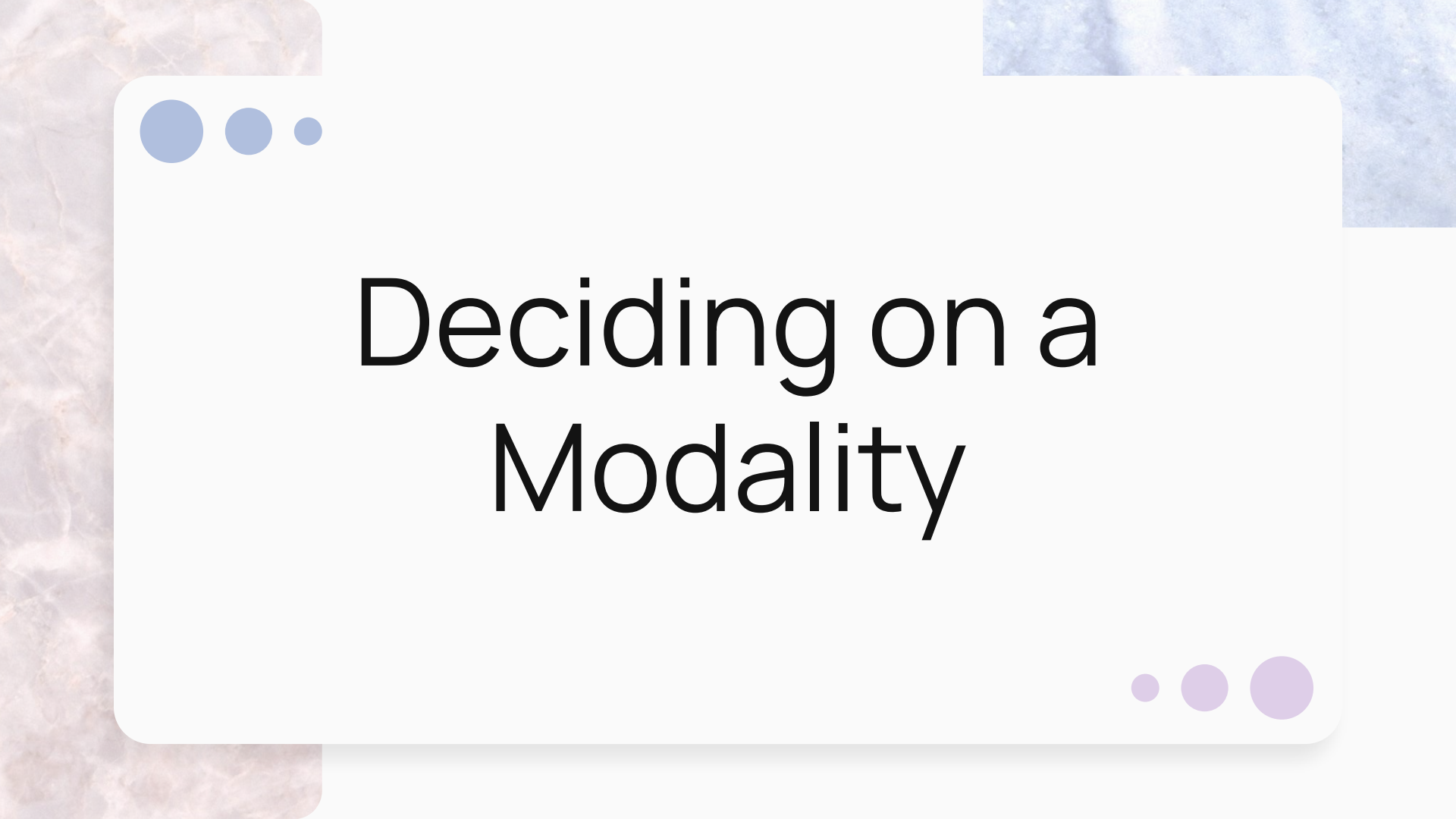


Barebones Phone App



AR Glasses





Deciding on a Modality



Barebones Phone App

Pros:

- Approximately 94 to 98% of Gen Z (our target audience) owns smartphones, which means our app will be accessible to more people.
- The fact that a smartphone has a screen makes it better to showcase products from around the world.
- Using smartphones is better suited to online shoppers because of the pre-existing apps that use this modality.
- Smartphones have been around longer, meaning that they have better security features than other devices. Some of our interviewees mentioned online safety as a big factor.

Cons:

- You actually need to hold the phone. This takes away from your hands being able to do something else in the meantime (some of our interviewees value being able to use hands to feel products, for example).
- Typing with your fingers can get repetitive and cause strain, especially for such a prompt-and-response app.
- Screens contribute to eye damage. There's a rising movement to avoid staying on screens for long periods, which may alienate some users.
- Smaller screen sizes limit the amount of information visible at once, which can make reviewing info less efficient than on larger devices.



AR Glasses

Pros:

- A big factor in getting the product that feels right is seeing how it looks. Many of our interviewees value looks and are often disappointed with how products bought online turn out. AR glasses offer a more realistic view of how a product will actually look.
- Using AR glasses keeps your hand free to actually feel the product, move a shopping cart, etc.
- The voice features of AR Glasses are quiet, maintaining privacy when shopping around others.
- AR overlays can add contextual information (price, reviews, size comparisons) directly into the user's field of view, reducing the need to switch between apps or screens.

Cons:

- We intended our application to suit online shopping, whereas AR glasses are more useful when you can visually scan products in a real store.
- Only approximately 25% of Gen Z users have AR glasses, so we wouldn't be very accessible.
- Many users often feel embarrassed when using the voice features of AR glasses, since it seems to others like they are talking to themselves.
- Current AR glasses are often expensive, bulky, or uncomfortable for long sessions, which may discourage extended shopping use.



Our Decision: Phone App with Voice Integration

We value accessibility and convenience. Ultimately, since almost everyone in our target audience of Gen Z audience already own phones, is is the obvious choice to consider. Many of the other modalities we considered (such as Google Home) would still require a phone to show images, so it doesn't make sense to use these other modalities. Although there are concerns about finger strain and eye strain, ultimately, the fact that Gen Z users are used to using phones for other shopping apps makes it most prudent for us to focus on phones as well.

However, we do value the conversational convenience of using Google Home. We believe that there is value in having voice as an option, so we will have a “talk using voice” button on our phone app to incorporate this other modality.

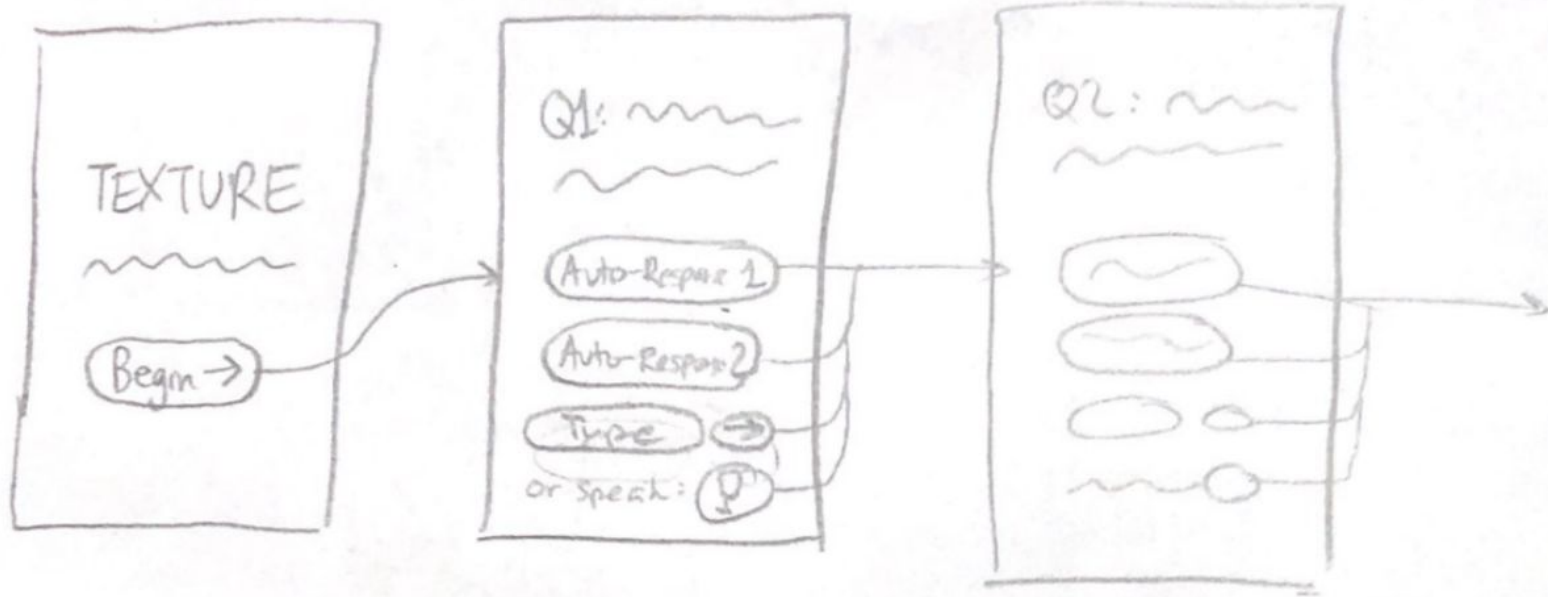




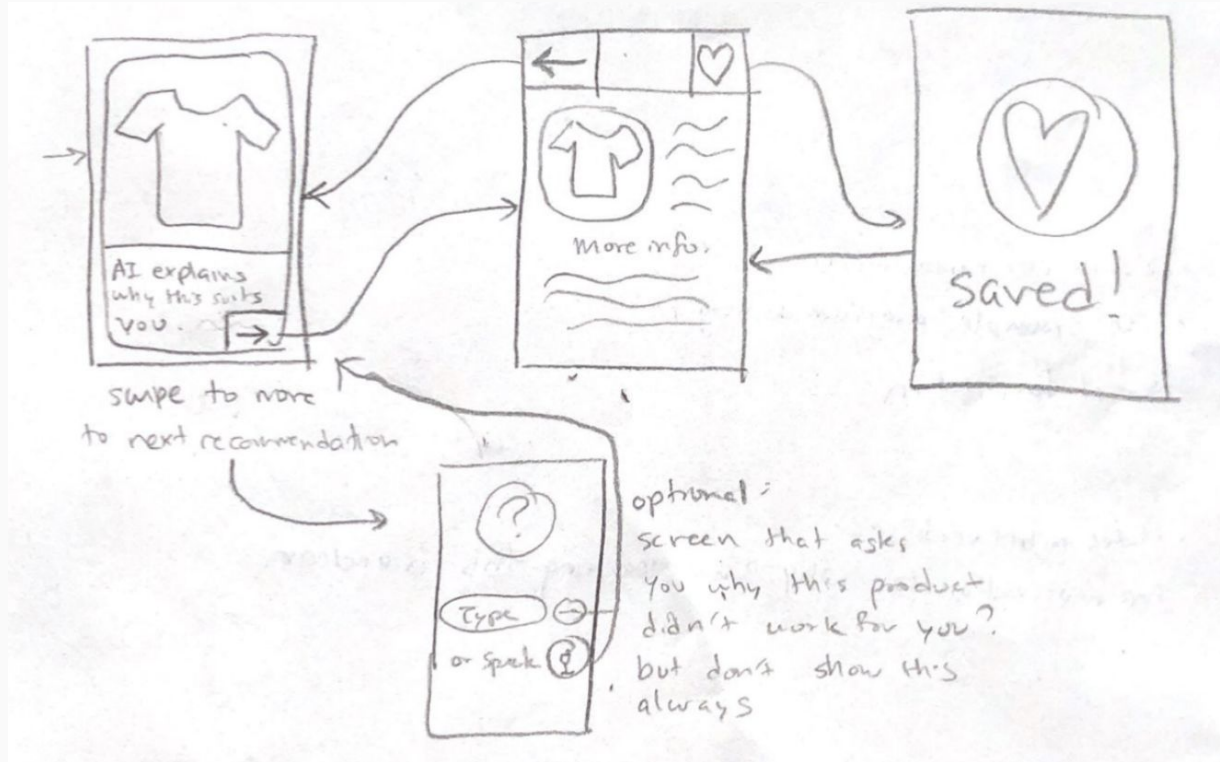
Task Flow Sketches



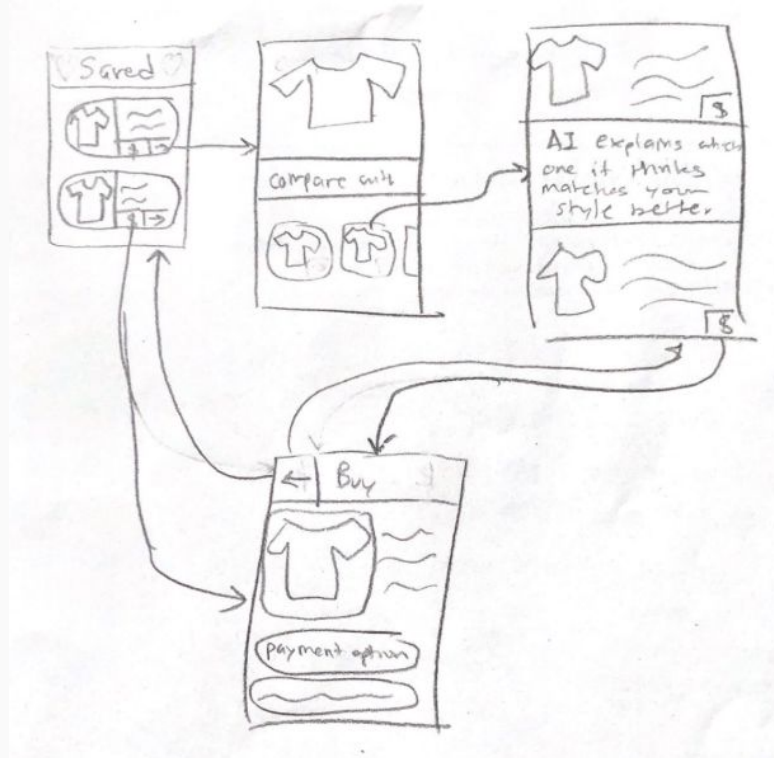
Simple Task: Expressing the User's Mood



Moderate Task: Searching for Products



Complex Task: Ask AI to compare items, reinforce purchases.





Low-Fi Prototype



How it was constructed?

Design:

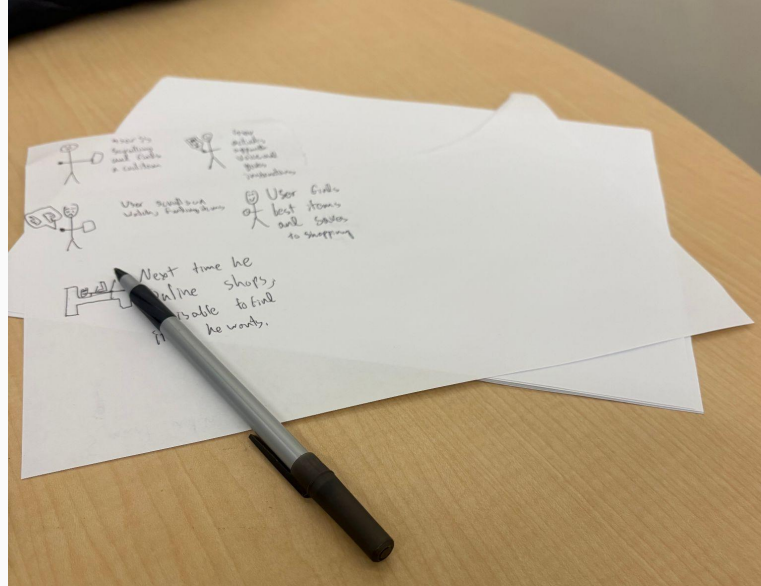
- We utilized pen and paper to allow for clear UI examples.
- We decided to use minimal words, placing importance on our titles.
- Multiple attempts were made until the team agreed with the prototype.

Features:

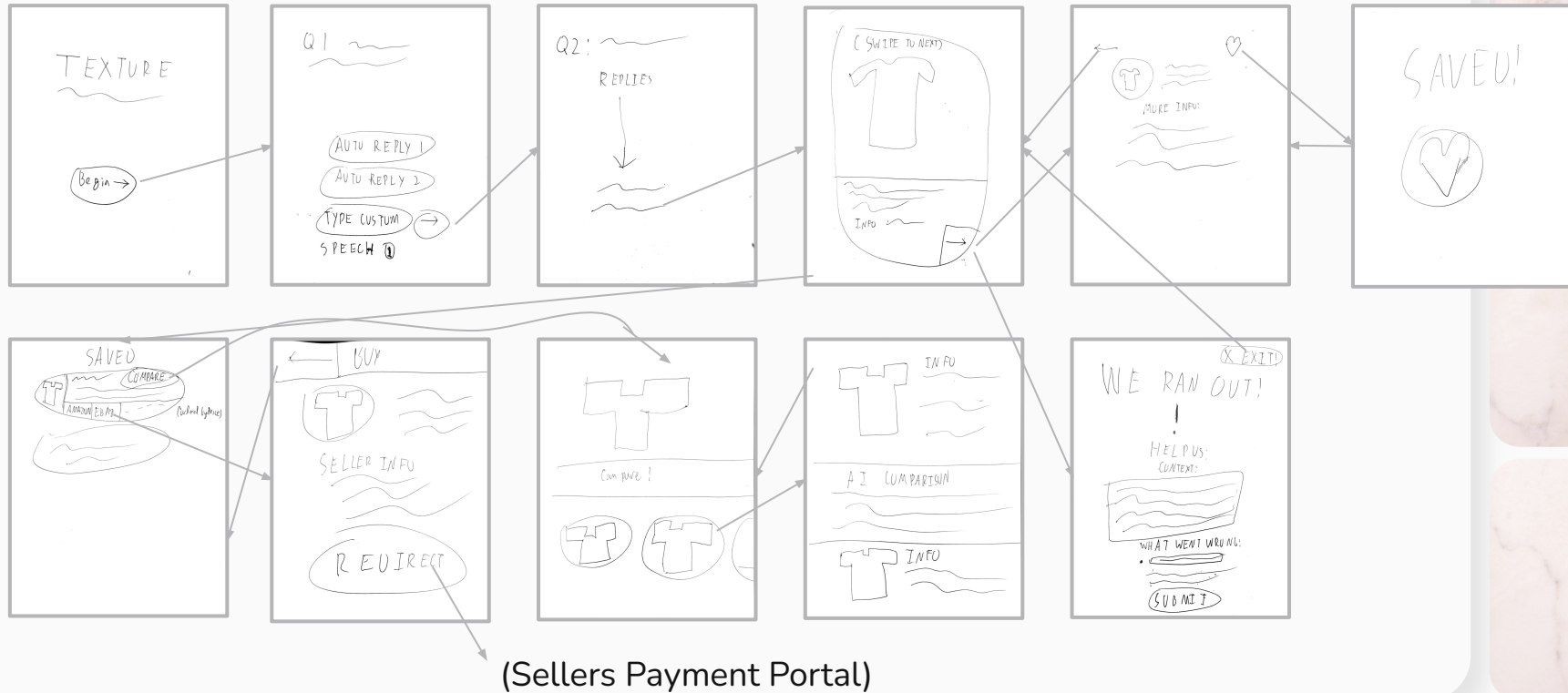
- Different options for the user to engage with suggested content
- Voice activation for quicker question answering.
 - For each question we have an option to record an answer.
- Skip buttons to streamline process for busy users.

Interactions:

- We had pages for each frame in our app, and we switched them as the user navigates. Our back buttons went back to the home screen and our save button showed a “Saved” screen popup.



Frames





User Testing



Testing Methodology

We were able to get people to interview with by exploring the streets of Isla Vista. Many of our target audience were either walking the streets of Del Playa or cooking food outside. We often walked up to people who were relaxing or finishing up their food, and asked if we could interview them.

Environment and Apparatus

USER 1

Location: Park

Equipment: Camera and Prototype



USER 2

Location: Dining Commons

Equipment: Camera and Prototype



USER 3

Location: House Party

Equipment: Camera and Prototype

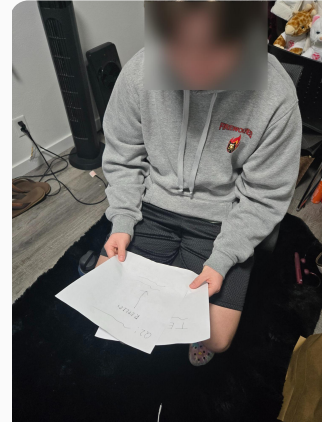


Testing Procedure

1. Introduce ourselves and the area we are researching in our app.
2. User is then prompted to start the application and go to the home app.
3. Ask questions to the interviewer about how he feels at each step, anything distracting or irritating?
4. Identify how long the user takes on each screen, noting when the user thinks or stalls.
5. Compare current experience to traditional one in Amazon.
6. Ask for the user to give any feedback on the current app flow.

Roles

Notetaker: Daniil and Ved
Prototype Handler: Rishik



Usability Goals / Requirements

Less Thinking Involved

- Average time spent on each frame
 - If a frame is confusing it might take for the user to process.
- Facial expressions the user experiences
 - If a user is feeling frustrated, he might make faces or exhibit negative emotions.

Impact:

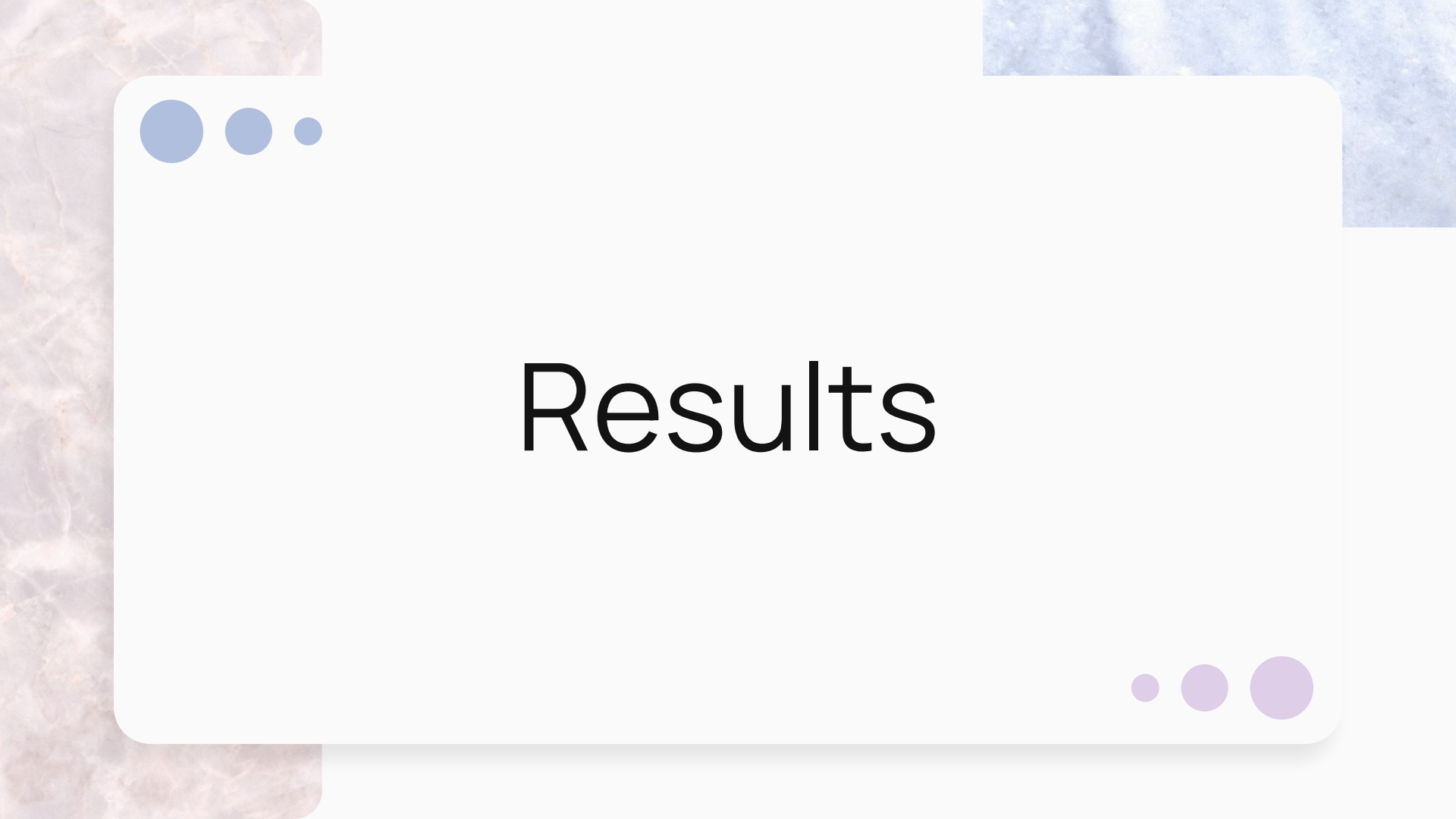
If our app doesn't strain the users mind like regular shopping, the user is more tempted to come back and use it again. We want the user to be alleviated of the stress of manually inspecting and comparing each item he finds while online shopping.

User Friendly UI

- The amount of mistakes the user makes in a frame.
 - If a user makes a mistake, he is probably unaware of the frames UI, and we need to redesign it on our part.

Impact:

If our app is swift and easy to use, the user is more tempted to come back and use it again. We want the user to be alleviated of the stress of manually inspecting and comparing each item he finds while online shopping.



Results



User 1

Information about participant: Community College student in Goleta we found at the park who occasionally shops online but when he does, he is very thorough in how he shops and always skeptical/review focused.

Feedback:

- Comparing products, add a filter based on comparing by price, quality, etc.
- Comparisons provide an overview of all reviews, which summarizes users' reviews
- Similarity ranking on how much a product is similar to the other/overlap.
- Likes the format of choosing products you like (swiping left or right on products)
- Likes the idea of the AI getting a feel for what you want, but is concerned about getting the recommendations for the same products over and over again, even after what the user's feeling/what they want changes.

User 2

Information about participant: Avid shopper, likes to spend money on high end branded products, met in a college dining commons.

Feedback:

- AI should keep track of budgeting maybe by connecting it to a spendings app or bank acc.
- Liked: Opening/landing page has a “how are we feeling today?” question that lets the user pick what kind of shopping experience they want. Examples: More straight forward shopping vs. browsing for fun/ emotional shopping in which you don’t want to spend too much money.
- Web extension when I am browsing on google, safari, etc.
- Liked: Tinder style shopping
- Would be fun if AI recommended things in a meme format
- Are you trying to help people buy more products and help the sellers or help people spend less time and money on products (helping the consumer be responsible)?

User 3

Information about Participant: met outside of an apartment house party in isla vista. Recent grad working in marketing, shops online frequently for clothes and tech accessories. Enjoys discovering new brands but buys on impulse sometimes.

Feedback:

- Liked the conversational AI idea, feels more engaging than scrolling endlessly on traditional shopping apps.
- Really liked the Tinder-style swiping format and said it makes finding thing feel fast and easy.
- Concerned about whether the AI would actually understand his mood or just push trending/popular items
- Suggested having a reset or new vibe button if he wants completely different recommendations
- Wondered if the app is encouraging impulse buying or helping users make smarter choices and thinks the mission should be clearer
- Would like quick digestible comparison summaries like pros/cons in 3 or 4 notes instead of long explanations

Statistics

Participant	Mistake s	Avg. Time Spent on Frame	Most Time Costly Frame
User 1 (Park)	1	19s	Product Comparison
User 2 (Dining)	0	6s	Q1: "How are you feeling?"
User 3 (Party)	2	13s	Product Comparison
AVERAGE	1	12.6s	--

Incidents

Both User 1 and User 3 got stuck on the comparison screen. They were unable to discern the functionality of our prototype.

During our interview, both User 3 and User 1 seemed slightly distant. User 1 clearly was trying to relax after a long week of midterms or classes. User 3 on the other hand was clearly in “party” mode, and very distracted.

User 2 seemed very attentive, already being on their phone. They were able to navigate through the app quickly and understand the functionality.



Implications and Redesign

- AI Echo Chamber
 - Users 1 and 3 expressed concern over the AI system, stating that it could give them the same vibe continuously.
 - **Resolution:** We need some sort of reset mechanism, to reset the AI so it could give more novel recommendations
- Clarity
 - Users 2 and 3 stated they were unsure of the Apps intentions: impulsive shopping or smart shopping? User 3 also asked for digestible comparisons.
 - **Resolution:** It is important that we add some sort of similarity score for each item, and a list of pros and cons.



Limitations

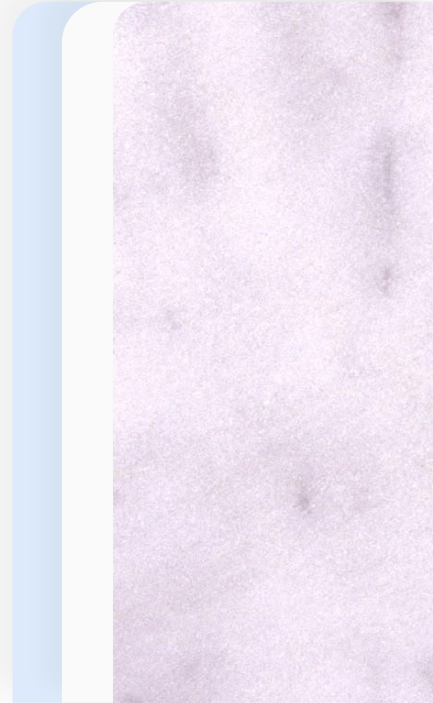
- AI Cant detect Emotions
 - There is no concrete way an AI can actually detect emotions. Its impossible to truly verify if a user is feeling what the AI is describing, which could lead to frustration.
- Personalization
 - We can't truly test whether our system becomes repetitive over time. For example, User 1 expressed this concern and we could only give her a single test to work on.
- Trust
 - We are unable to truly test whether the user trusts the platform they were recommended by our app, as we stop the simulation at the buy button.
- Testing Environment
 - We noticed that people who were relaxing in a park or in a house party on average were less attentive to our app, adding in a factor of variability in our experiments.

Appendix:

Sketching Report

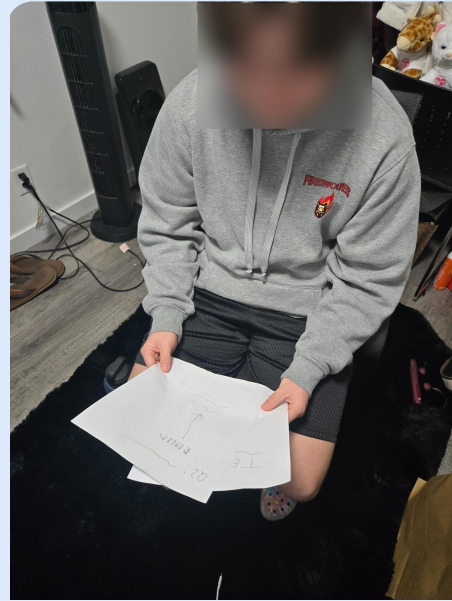
Here is a link to the complete sketching report:

<https://docs.google.com/document/d/1ukVZFGIngQKChpzQlkH0B30khGoOSDXvEQvFSD0v1RM/edit?usp=sharing>



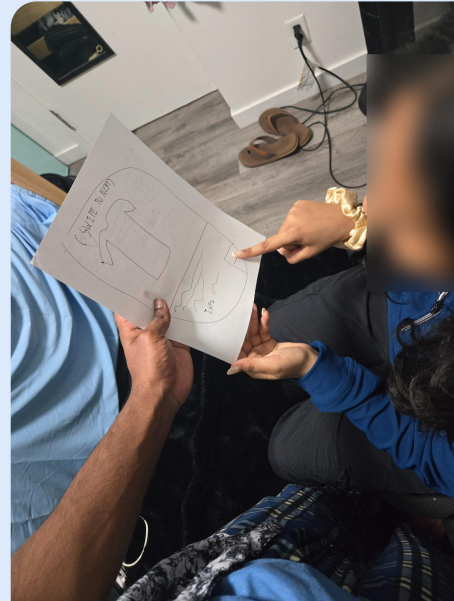
Appendix:

User Testing Pictures



Appendix:

User Testing Pictures





Thank you!